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Socket head cap screw dimensions: metric ISO 4762, inch ASME B18.3, property classes

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1. ISO 4762 socket head cap screw dimensions, M1.6 to M36

Dimensions in millimetres, maximum values for the head (ISO 4762, 2004 edition). P: thread pitch (coarse). dk: head diameter. k: head height. s: hex key size, i.e. the socket width across flats. t: minimum socket depth. b: reference thread length of partially threaded screws.

Screw	P	Head Ø dk	Height k	Key s	Socket depth t min	Thread b ref
M1.6	0.35	**3**	**1.6**	**1.5**	0.7	15
M2	0.4	**3.8**	**2**	**1.5**	1	16
M2.5	0.45	**4.5**	**2.5**	**2**	1.1	17
M3	0.5	**5.5**	**3**	**2.5**	1.3	18
M4	0.7	**7**	**4**	**3**	2	20
M5	0.8	**8.5**	**5**	**4**	2.5	22
M6	1	**10**	**6**	**5**	3	24
M8	1.25	**13**	**8**	**6**	4	28
M10	1.5	**16**	**10**	**8**	5	32
M12	1.75	**18**	**12**	**10**	6	36
M14 (non-preferred)	2	**21**	**14**	**12**	7	40
M16	2	**24**	**16**	**14**	8	44
M20	2.5	**30**	**20**	**17**	10	52
M24	3	**36**	**24**	**19**	12	60
M30	3.5	**45**	**30**	**22**	15.5	72
M36	4	**54**	**36**	**27**	19	84

Head diameters are for a plain head; a knurled head may be slightly larger (10.22 mm instead of 10 mm for M6). The standard goes up to M64. Short screws are threaded up to the head: the limit depends on the size, see the standard or the supplier's catalogue.

2. Inch socket head cap screws (ASME B18.3)

Inch-series socket head cap screws, UNC thread: 1/4-20 reads "1/4 inch diameter, 20 threads per inch". Head: maximum values in inches (mm equivalent in brackets, 1 inch = 25.4 mm). As in metric, the head height equals the diameter, and the head diameter is 1.5 times the diameter from 1/4 up.

Screw (UNC)	Head Ø A (in)	Height H (in)	Hex key J	Key engagement T min (in)
#4-40	**0.183** (4.65 mm)	**0.112** (2.84 mm)	**3/32"** (2.38 mm)	0.051
#5-40	**0.205** (5.21 mm)	**0.125** (3.18 mm)	**3/32"** (2.38 mm)	0.057
#6-32	**0.226** (5.74 mm)	**0.138** (3.51 mm)	**7/64"** (2.78 mm)	0.064
#8-32	**0.270** (6.86 mm)	**0.164** (4.17 mm)	**9/64"** (3.57 mm)	0.077
#10-24	**0.312** (7.92 mm)	**0.190** (4.83 mm)	**5/32"** (3.97 mm)	0.090
1/4-20	**0.375** (9.53 mm)	**0.250** (6.35 mm)	**3/16"** (4.76 mm)	0.120
5/16-18	**0.469** (11.91 mm)	**0.312** (7.92 mm)	**1/4"** (6.35 mm)	0.151
3/8-16	**0.562** (14.27 mm)	**0.375** (9.53 mm)	**5/16"** (7.94 mm)	0.182

Screw (UNC)	Head Ø A (in)	Height H (in)	Hex key J	Key engagement T min (in)
7/16-14	**0.656** (16.66 mm)	**0.438** (11.13 mm)	**3/8"** (9.53 mm)	0.213
1/2-13	**0.750** (19.05 mm)	**0.500** (12.70 mm)	**3/8"** (9.53 mm)	0.245
5/8-11	**0.938** (23.83 mm)	**0.625** (15.88 mm)	**1/2"** (12.70 mm)	0.307
3/4-10	**1.125** (28.58 mm)	**0.750** (19.05 mm)	**5/8"** (15.88 mm)	0.370
7/8-9	**1.312** (33.32 mm)	**0.875** (22.23 mm)	**3/4"** (19.05 mm)	0.432
1-8	**1.500** (38.10 mm)	**1.000** (25.40 mm)	**3/4"** (19.05 mm)	0.495

3. Property classes 8.8, 10.9, 12.9 (ISO 898-1)

Steel socket head cap screws come in property classes 8.8, 10.9 and 12.9. The first number gives the nominal tensile strength (12 → 1,200 MPa), the second the ratio of yield to tensile strength (.9 → 90%). ISO 898-1 values, in MPa:

Class	Diameter	Nominal R_m	Min R_m	Min $R_{p0.2}$
8.8	d ≤ 16 mm	800	**800**	**640**
8.8	d > 16 mm	800	**830**	**660**
10.9	all	1000	**1040**	**940**
12.9	all	1200	**1220**	**1100**

In stainless steel, socket head cap screws are most often A2-70 or A4-70 (ISO 3506-1): A2 or A4 is the austenitic grade, 70 a tensile strength of at least 700 MPa. The tightening torque depends on the class and on friction: see the bolt torque chart.

Reference values taken from the article https://www.baouch.fr/en/blog/vis-chc-iso-4762-dimensions/, which explains the assumptions, formulas and edge cases. For critical use, check the current standard or the manufacturer's documentation. Free to print and share with credit to the source.